

Resource Library: Neighborhood Sharing System

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INTRODUCTION

In today's world, many household tools and resources are used only occasionally, leading to unnecessary purchases and wastage. Instead of buying items that are rarely used, sharing them within a community can be more efficient and cost-effective. This project introduces a Resource Library system that enables users to share and borrow items within a neighborhood. The system aims to promote collaboration, reduce expenses, and encourage sustainable usage of resources.

PROBLEM STATEMENT

There is currently no simple and organized platform for sharing household tools within local communities. Many items remain unused most of the time, while others may need them temporarily. Traditional methods of borrowing are informal and lack proper tracking, which can lead to confusion, delays, or loss of items. Therefore, there is a need for a digital system that can manage item availability, borrowing, and returning efficiently.

ABSTRACT

The Resource Library is a web-based application designed to facilitate peer-to-peer sharing of household items. It allows users to add items, borrow them, and return them while maintaining a clear record of their status. The system tracks the lifecycle of each item, ensuring that users know whether an item is available or borrowed. Additional features such as borrower details and due dates improve accountability and usability. The application is built using modern web technologies, ensuring a simple yet effective user experience.

OBJECTIVE

The main objective of this project is to develop a user-friendly system that enables efficient sharing of resources within a community. It aims to allow users to add items, view available items, borrow items, and return them seamlessly. The system also focuses on tracking the lifecycle of items and storing all relevant data in a structured manner. Ultimately, the goal is to create a scalable solution that promotes resource optimization and community collaboration.

SYSTEM FEATURES

The system provides essential features required for managing shared resources. Users can add new items along with their details, view all available items, and borrow items by providing borrower information and due dates. The system updates the status of items dynamically, indicating whether they are available or borrowed. It also allows users to return items, automatically resetting their status. A simple and clean user interface ensures ease of use.

TECHNOLOGIES USED

This project is developed using a combination of frontend and backend technologies. The backend is built using Spring Boot, which provides RESTful APIs for handling business logic and database operations. The frontend is developed using React.js, enabling a responsive and interactive user interface. MySQL is used as the database to store item details. Tools such as Maven, Eclipse IDE, and Visual Studio Code are used for development and project management.

SYSTEM ARCHITECTURE

The system follows a three-tier architecture consisting of the frontend, backend, and database. The React frontend interacts with the Spring Boot backend through REST APIs to send and receive data. The backend processes requests, applies business logic, and communicates with the MySQL database to store and retrieve information. This architecture ensures a clear separation of concerns and improves scalability and maintainability.

WORKFLOW

The workflow of the system is simple and intuitive. A user first adds an item to the system, which is then stored in the database with an “Available” status. When another user borrows the item, the system updates its status to “Borrowed” and records the borrower’s details along with the due date. Once the item is returned, its status is updated back to “Available.” This cycle continues, ensuring proper tracking of each item.

IMPLEMENTATION

The implementation involves developing REST APIs using Spring Boot to handle operations such as adding, retrieving, borrowing, and returning items. The frontend is connected to these APIs using Axios, enabling smooth communication between the user interface and backend. The database is integrated using JPA and Hibernate, which automatically manage table creation and data persistence. The user interface is enhanced using CSS to provide a clean and modern look.

CONCLUSION

The Resource Library system successfully demonstrates how technology can be used to facilitate efficient sharing of resources within a community. It provides a simple yet effective solution for managing item availability and tracking their lifecycle. With further improvements, the system has the potential to evolve into a full-scale platform for resource sharing.

Output:

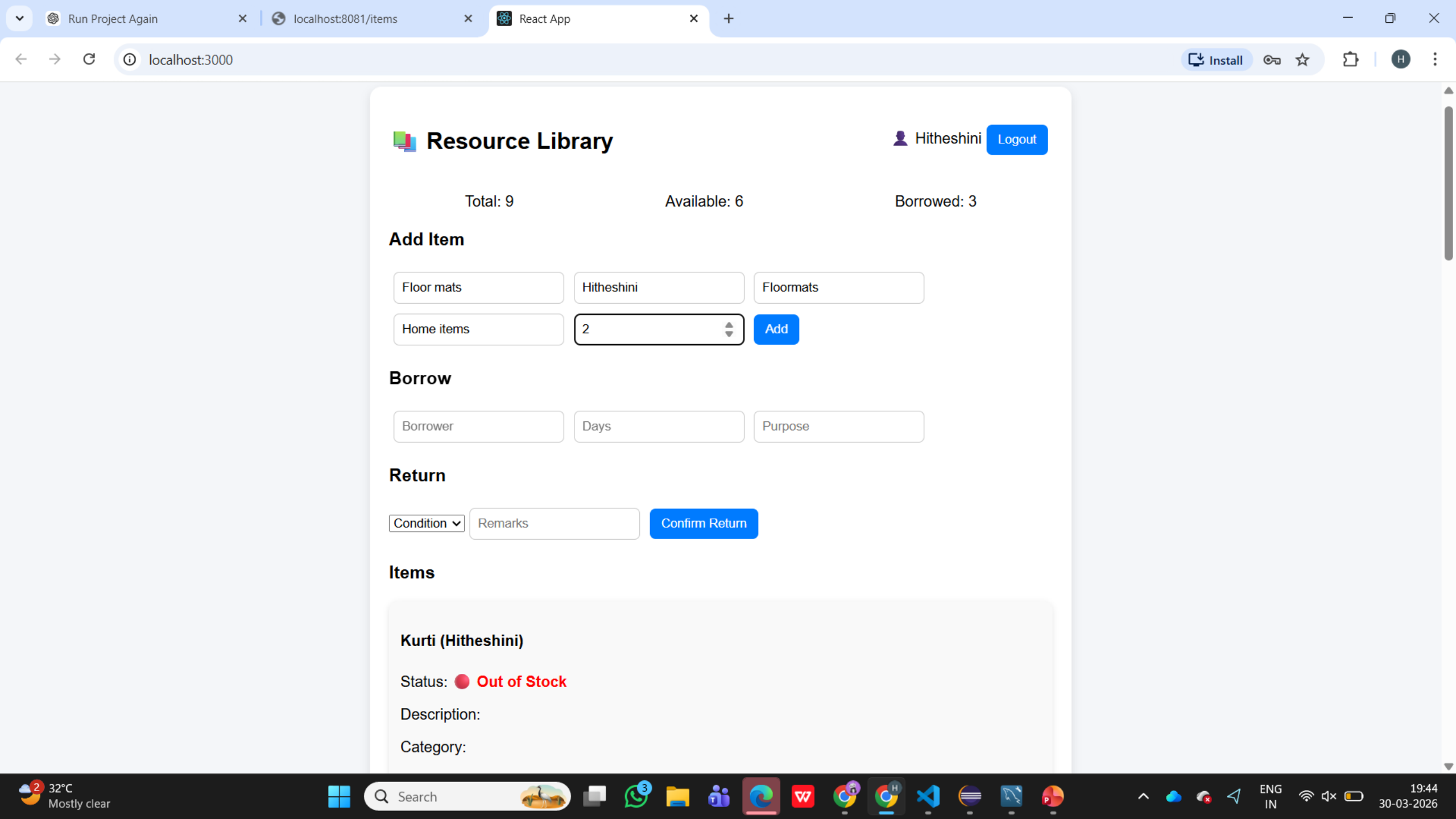
Login

Hitheshini

....

Login

[New user? Signup](#)



Resource Library

Hitheshini [Logout](#)

Total: 9

Available: 6

Borrowed: 3

Add Item

Floor mats	Hitheshini	Floormats
Home items	2	Add

Borrow

Borrower	Days	Purpose
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Return

Condition	Remarks	Confirm Return
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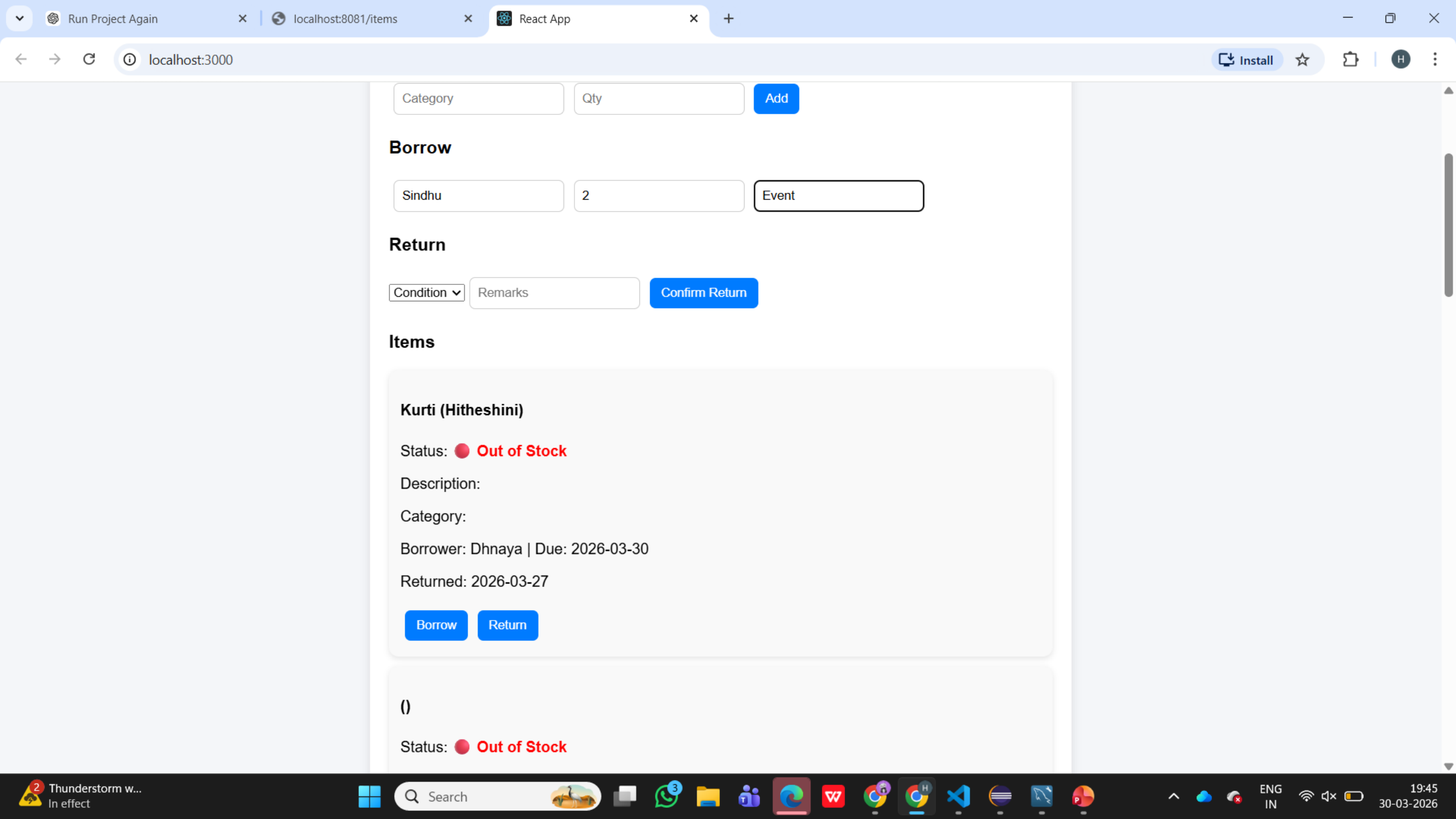
Items

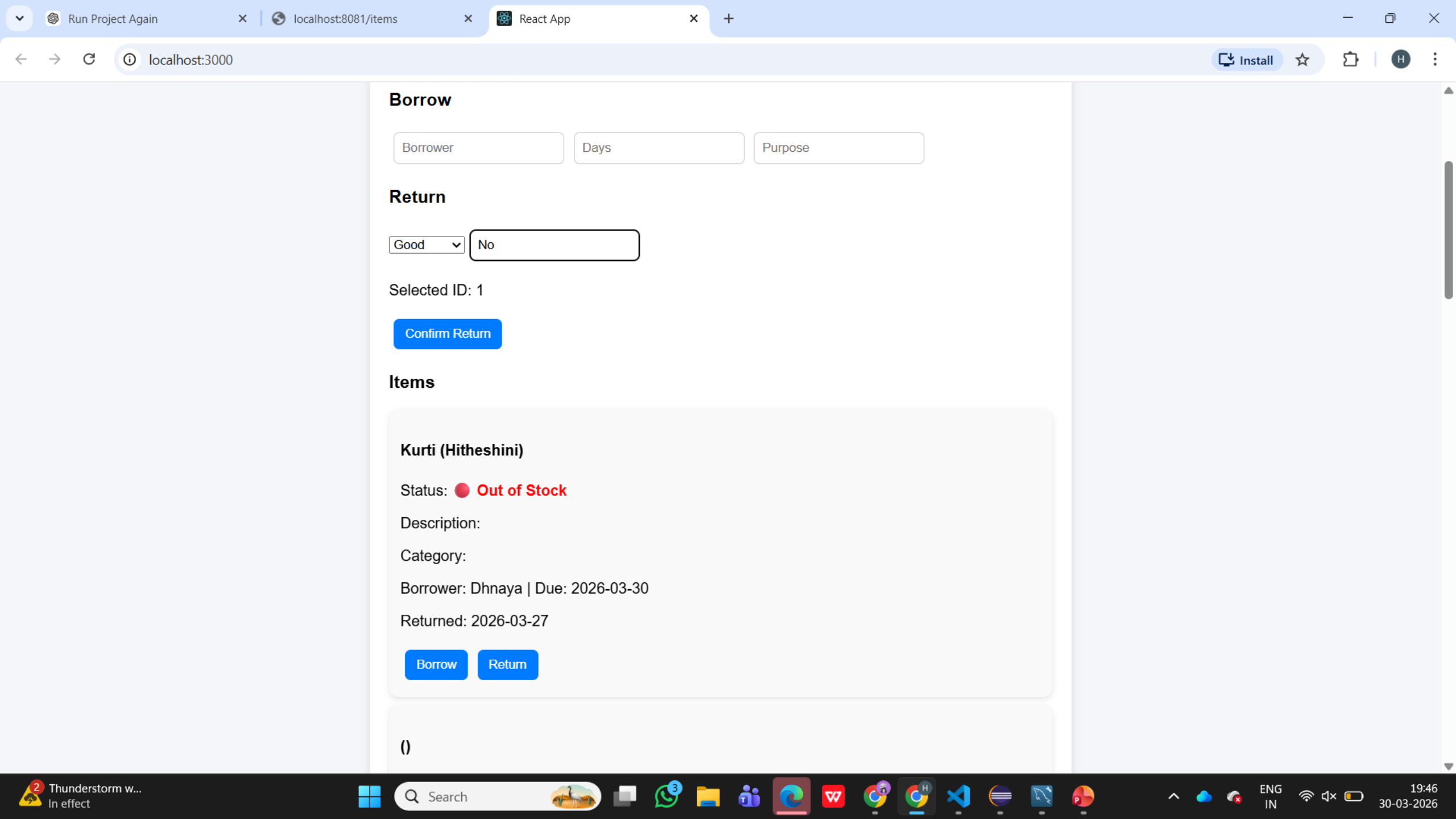
Kurti (Hitheshini)

Status: ● Out of Stock

Description:

Category:





Borrow

Return

Selected ID: 1

Confirm Return

Items

Kurti (Hitheshini)

Status: ● Out of Stock

Description:

Category:

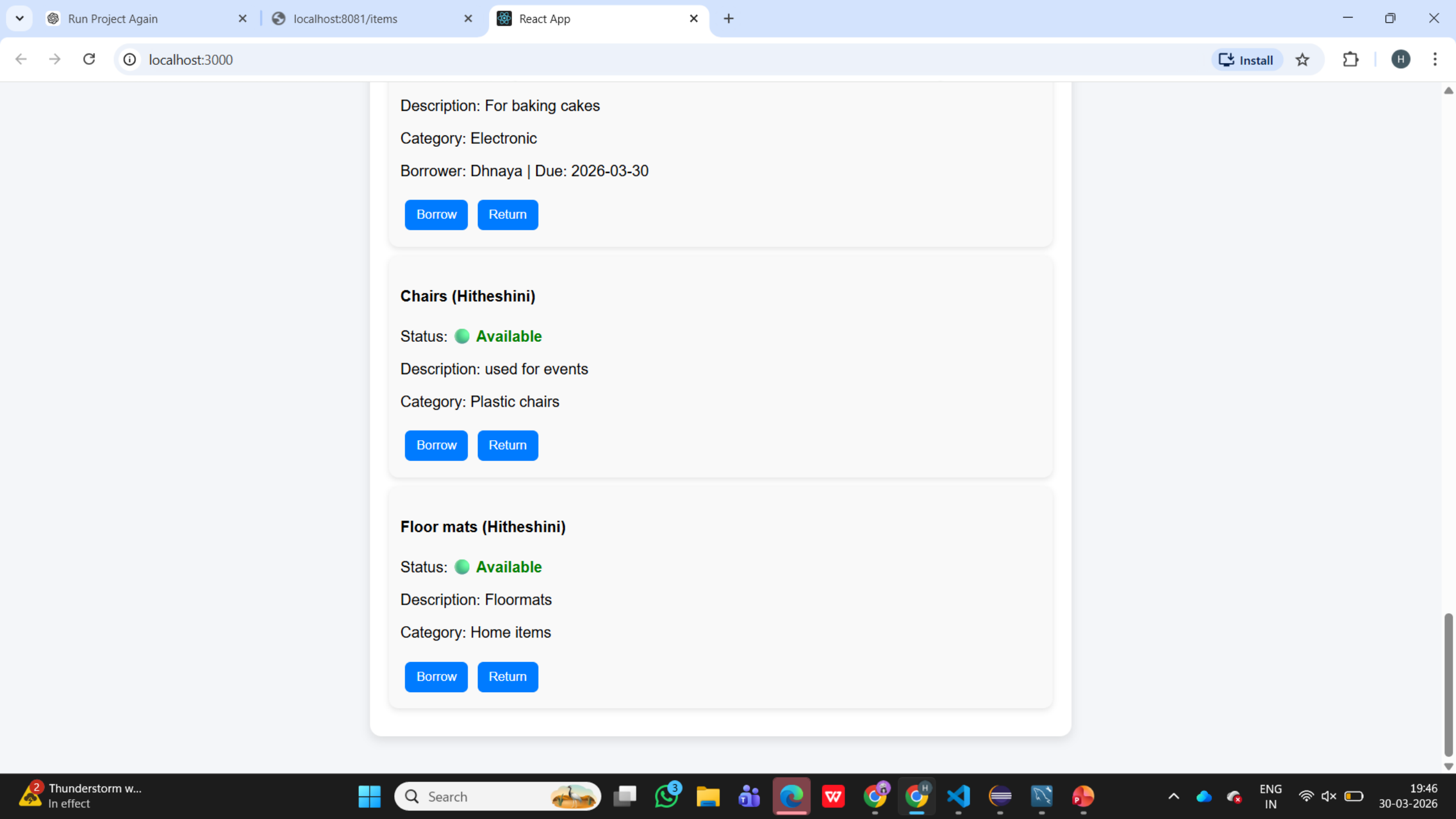
Borrower: Dhnaya | Due: 2026-03-30

Returned: 2026-03-27

Borrow

Return

()



Description: For baking cakes

Category: Electronic

Borrower: Dhnaya | Due: 2026-03-30

Borrow

Return

Chairs (Hitheshini)

Status: ● Available

Description: used for events

Category: Plastic chairs

Borrow

Return

Floor mats (Hitheshini)

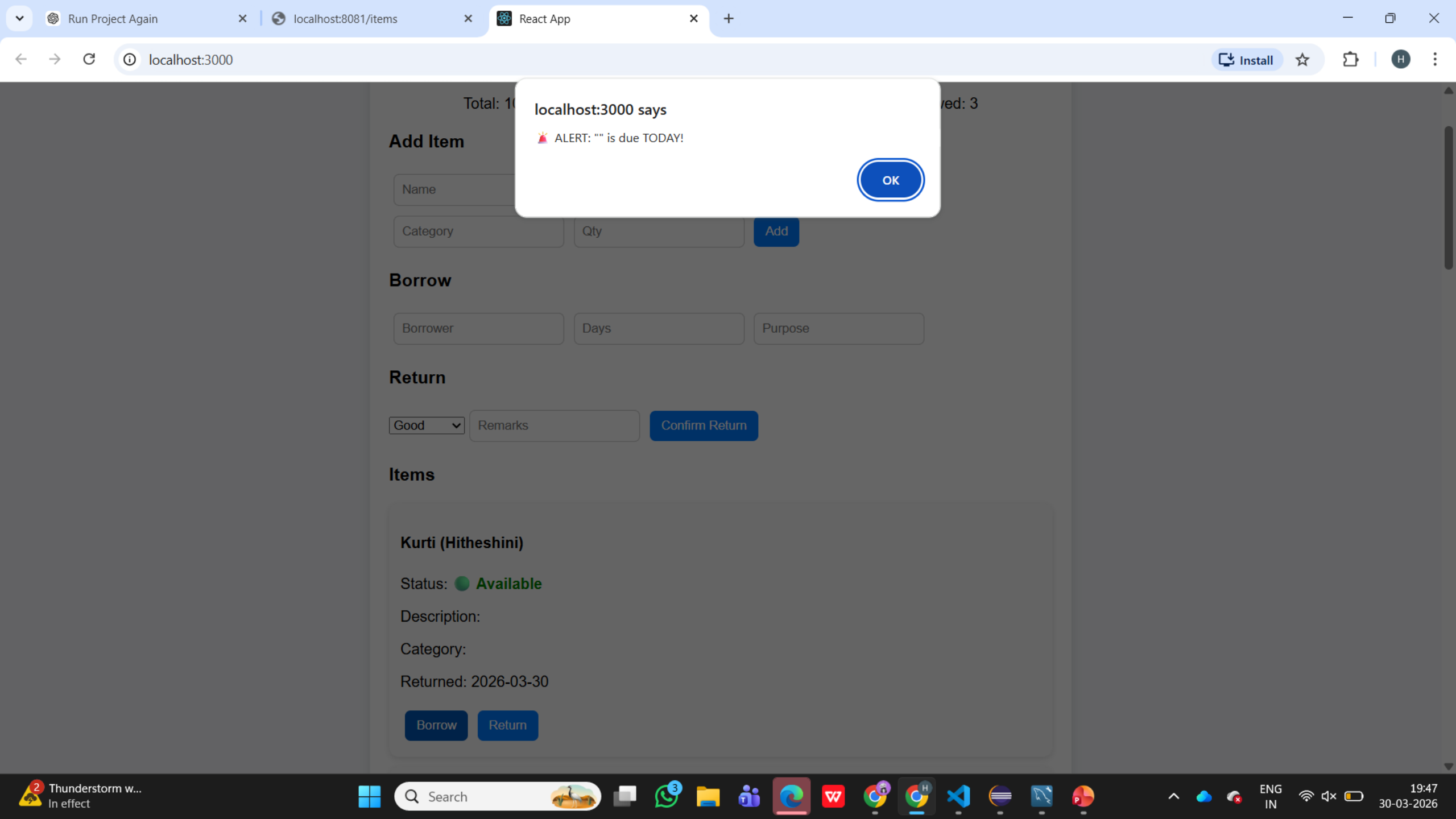
Status: ● Available

Description: Floormats

Category: Home items

Borrow

Return



localhost:3000 says

🔔 ALERT: "" is due TODAY!

OK

Add Item

Add

Borrow

Return

Good

Confirm Return

Items

Kurti (Hitheshini)

Status: ● Available

Description:

Category:

Returned: 2026-03-30

Borrow

Return